OSL hardware–simulation comparison architecture

Hardware host / Raspberry Pi

Telemetry HTTP server

Bounded in-memory sample buffer
knee_telemetry.py; no database

Sample publish

Reference generator + physical knee PID

Target and encoder error → current command
Nominal control rate: 200 Hz

Motor driver + physical drivetrain

Current command over CAN
Motor / transmission / belt

Physical knee + joint encoder

Actual knee angle via AS5048B / I²C
Tested setup: isolated knee fixture

Encoder feedback

Measured angle / validity

Wi-Fi / IP

HTTP / TCP :8794
JSON batches; token

Simulation computer

Receiver + source-time buffer

HTTP polling → sequence / validity checks
Save raw data; default live buffer: 100 ms

Target + source time

Independent simulation PID

Same target; own simulated feedback
200 Hz; C or knee-error derivative

Current conversion + drivetrain model

Current → actuator dynamics / friction
Elastic belt → knee torque

MuJoCo rigid-body plant

Physics timestep: 0.5 ms (2 kHz)
Full-leg or separate knee-only adapter

Simulated feedback

Recorded hardware samples

Time-aligned comparison, plots and saved records

Hardware angle + simulated angle + target; tracking errors and RMS / peak differences
Simulation currents, torques, belt deflection; CSV / JSON records

Timing: source timestamps drive multiple fixed physics steps per sample (~10 at 200 Hz); no Pi–simulation handshake.

Replay: saved samples replace the network input. Model calibration is a separate offline activity.

Measured encoder angles initialize the simulation once and provide comparison data; subsequent control uses simulated state.